Commutative Algebra

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Lecture 1

In this course, a ring is a commutative unital ring R. One non-commutative exception is the ring $\mathrm{End}(M)$, where M is an abelian group. This is a ring with pointwise addition, and composition as multiplication.

Definition 0.1 (module)

An *R-module* M is an abelian group M with an fixed ring homomorphism $\rho : R \to \mathrm{End}(M)$. We will write $r \cdot m := \rho(r)(m)$.

Remark 0.2. By definition, this implies that $(r_1 + r_2) \cdot m = r_1 \cdot m + r_2 \cdot m$, $r \cdot (m_1 + m_2) = r \cdot m_1 + r \cdot m_2$ and $r_1 \cdot (r_2 \cdot m) = (r_1 r_2) \cdot m$.

Example 0.3 (Examples of modules)

- Let k be a field. Then a k-module is the same as a k-vector space.
- Every abelian group is a $\mathbb{Z}$-module in a unique way, since we must have that $\rho(1) = \mathrm{id}_M$. Therefore, abelian groups and $\mathbb{Z}$-modules are the same thing.
- Every ring R is (trivially) an R-module.
- More generally, $R^{\oplus \mathbb{N}}$ (direct sum) and $R^{\mathbb{N}}$ (direct product) are R-modules.

1 Chain conditions

Definition 1.1 (Noetherian, Artinian module)

An R-module M is *Noetherian* if one of the following (equivalent) conditions hold:

1. Every ascending chain of submodules $M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots$ stabilises. That is, it is eventually constant.
2. Every non-empty set Σ of submodules of M has a maximal element.

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M is *Artinian* if we replace in the above: ascending with descending, maximal with minimal.

Lemma 1.2. An R-module M is Noetherian if and only if every submodule of M is finitely generated.

In particular, every Noetherian module is finitely generated. If $R = \mathbb{Z}[T_1, T_2, \dots]$, with $M = R$ as an R-module. Then M is finitely generated. On the other hand, $M' = \langle T_1, T_2, T_3, \dots \rangle$, is not finitely generated.

Definition 1.3 (Noetherian, Artinian ring)
A ring R is Noetherian (resp. Artinian) if it is Noetherian (resp. Artinian) as an R-module.

Example 1.4
1. $\mathbb{Z}$ is Noetherian (as it is a PID), but not Artinian (e.g. $\langle 2 \rangle \supseteq \langle 4 \rangle \supseteq \langle 8 \rangle \supseteq \cdots$).
2. $\mathbb{Z}[1/2]/\mathbb{Z}$ is Artinian, but not Noetherian as a $\mathbb{Z}$-module.
3. A ring R is Artinian if and only if R is Noetherian and R has Krull dimension 0.

Definition 1.5 (Exact sequence)
A sequence

$$\cdots \longrightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \longrightarrow \cdots$$

of R-modules and R-module homomorphisms is *exact* if $\operatorname{im}(f_i) = \ker(f_{i+1})$ for all i.

Definition 1.6 (Short exact sequence)
A *short exact sequence* (SES) is an exact sequence of the form

$$0 \longrightarrow N \xhookrightarrow{\iota} M \twoheadrightarrow L \longrightarrow 0$$

That is, we have an embedding $\iota : N \hookrightarrow M$, and an isomorphism $L \cong M/\iota(N)$.

Lemma 1.7. Let

$$0 \longrightarrow N \longrightarrow M \longrightarrow L \longrightarrow 0$$

be an SES of R-modules. Then M is Noetherian (resp. Artinian) if and only if N and L are Noetherian (resp. Artinian).

Proof. We may assume without loss of generality that N is a submodule of M. Let $P_1 \subseteq P_2 \subseteq \dots$ be an increasing (resp. decreasing) sequence of submodules of M. In this case,

$$N \cap P_1 \subseteq N \cap P_2 \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of N, hence eventually constant. Similarly,

$$\frac{N + P_1}{N} \subseteq \frac{N + P_2}{N} \subseteq \cdots$$

is an increasing (resp. decreasing) sequence of submodules of $L = M/N$, hence eventually constant. For large n, we will have

$$P_n \subseteq P_{n+1} \quad N \cap P_n = N \cap P_{n+1} \quad N + P_n = N + P_{n+1}$$

Hence $P_n = P_{n+1}$ for large enough n. □

Corollary 1.8. If $M_1, \dots, M_n$ are Noetherian (resp. Artinian) R-modules, then $M_1 \oplus \cdots \oplus M_n$ is Noetherian (resp. Artinian).

Proof. By the lemma and induction. □

Recall a module homomorphism

$$\varphi : M_1 \oplus \cdots \oplus M_n \to L$$

is the same as a collection of module homomorphism $\varphi_i : M_i \to L$. This is also true for infinite direct sums (but not products!).

Proposition 1.9. For a Noetherian (resp. Artinian) ring R, every finitely generated R-module is Noetherian (resp. Artinian).

Proof. M is finitely generated if and only if there exists a surjection $R^n \twoheadrightarrow M$ for some $n \in \mathbb{N}$. The fact that R^n is Noetherian (resp. Artinian) implies that M is Noetherian (resp. Artinian), as quotients of Noetherian (resp. Artinian) modules are Noetherian (resp. Artinian). □

Definition 1.10 (algebra)

An *R-algebra* A is a ring A with a fixed ring homomorphism $\rho : R \to A$. We will write $r \cdot a := \rho(r)a$.

Definition 1.11 (noetherian algebra)

An R-algebra is *Noetherian* if it is Noetherian as a ring.

Remark 1.12. Every R-algebra is an R-module.

Example 1.13

The polynomial ring $k[T_1, \dots, T_n]$ is a k-algebra. Do note however that it is a finitely generated by $T_1, \dots, T_n$ as a k-algebra, but it is infinite dimensional as a k-vector space.

Definition 1.14 (algebra homomorphism)

$\varphi : A \to B$ is an *R-algebra homomorphism* if φ is a ring homomorphism and $\varphi(r \cdot 1_A) = r \cdot 1_B$.

Definition 1.15 (finitely generated algebra)

An R-algebra A is *finitely generated* if there exists a surjective R-algebra homomorphism $R[T_1, \dots, T_n] \twoheadrightarrow A$ for some $n \in \mathbb{N}$.

Lecture 2

Theorem 1.16 (Hilbert basis theorem)**.** Every finitely generated algebra A over a Noetherian ring R is Noetherian (as a ring).

For example, if k is a field, then $k[T_1, \dots, T_n]$ is Noetherian.

Proof. It suffices to prove for $A = R[T_1, \dots, T_n]$, since every finitely generated algebra is a quotient of $R[T_1, \dots, T_n]$. Moreover, by induction, suffices to prove the result for $A = R[T]$.

Let $\mathfrak{a}$ be an ideal of $A = R[T]$. For every $i \geq 0$, define

$$\mathfrak{a}(i) = \left\{ c_0 \mid c_0 t^i + \cdots + c_i t^0 \in \mathfrak{a} \right\}$$

for the set of all leading coeffients of elements of degree i in $\mathfrak{a}$ (and containing 0). In this case, $\mathfrak{a}(i) \subseteq R$ is an ideal, and we have an ascending chain of ideals

$$\mathfrak{a}(i) \subseteq \mathfrak{a}(i+1) \subseteq \cdots$$

Since R is Noetherian, each $\mathfrak{a}$ is finitely generated (as an ideal), and the ascending sequence of ideal stabilises. That is,

$$\mathfrak{a}(m') = \mathfrak{a}(m)$$

for all $m' \geq m$. We write $\mathfrak{a}(i) = \langle b_{i,1}, \dots, b_{i,m_i} \rangle$, where $b_{i,j} \in R$. Let $f_{i,j} \in \mathfrak{a}$ be a polynomial of degree i, with leading coefficient $b_{i,j}$. Define the new ideal

$$\mathfrak{b} = \left\langle f_{i,j} \mid i \leq m, 1 \leq j \leq m_i \right\rangle \trianglelefteq R[T]$$

In this case, $\mathfrak{b}(i) = \mathfrak{a}(i)$ for all i. By construction, $\mathfrak{b} \subseteq \mathfrak{a}$.

Suppose for contradiction that $\mathfrak{a} \nsubseteq \mathfrak{b}$. Take $f \in \mathfrak{a} \setminus \mathfrak{b}$ of minimal degree i. But $\mathfrak{b}(i) = \mathfrak{a}(i)$, and so there exists $g \in \mathfrak{b}$, of degree i, and with the same leading coefficient as f. That is, $\deg(f - g) < i$. By minimality, $f - g \in \mathfrak{b}$, and so $f = (f - g) + g \in \mathfrak{b}$. Contradiction. □

Therefore, if we have a subset $S \subseteq R[T_1, \dots, T_n]/I$, then $\langle S \rangle = \langle S_0 \rangle$, where $S_0 \subseteq S$ is finite.

2 Tensor products

Let M, N be R-modules. An informal definition of their tensor product is

$$M \otimes_R N = \left\{ \sum_{i=1}^{\ell} m_i \otimes n_i \;\middle|\; m_i \in M, n_i \in N \right\}$$

where we have the relations $(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n$, $m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2$, and that for $r \in R$, $(rm) \otimes n = r(m \otimes n) = m \otimes (rn)$.

For example, consider $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3$. Then

$$x \otimes y = (3x) \otimes y = x \otimes (3y) = x \otimes 0 = 0$$

and so, $\mathbb{Z}/2 \otimes_{\mathbb{Z}} \mathbb{Z}/3 = 0$. On the other hand, if we have vector spaces, then

$$\mathbb{R}^m \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong \mathbb{R}^{m\ell}$$

Recall $f : M \times N \to L$ is *R-bilinear* if $n \mapsto f(m_0, n)$ and $m \mapsto f(m, n_0)$ are R-linear for all $m_0 \in M, n_0 \in N$.

Definition 2.1 (tensor product of modules)

Let M, N be R-modules, let

$$\mathcal{F} = R^{\oplus(M \times N)} = \operatorname{span}_R \left\{ e_{(m,n)} \mid m \in m, n \in N \right\}$$

be the free module indexed by $m \times n$, and define $\mathcal{K} \subseteq \mathcal{F}$ for the submodule generated by the relations (where we write (m, n) for $e_{(m,n)}$)

$$\begin{aligned} (m, n_1) + (m, n_2) &= (m, n_1 + n_2) \\ (m_1, n) + (m_2, n) &= (m_1 + m_2, n) \\ r(m, n) &= (rm, n) \\ r(m, n) &= (m, rn) \end{aligned}$$

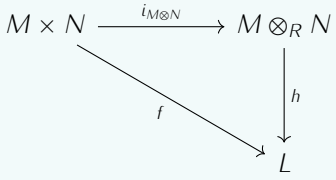
The *tensor product* is

$$M \otimes_R N := \frac{\mathcal{F}}{\mathcal{K}}$$

We have an R-bilinear map

$$\begin{aligned} i_{M\otimes N} : M \times N &\to M \otimes_R N \\ (m, n) &\mapsto m \otimes n \end{aligned}$$

Proposition 2.2 (universal property of tensor product)**.** For every R-module L and any R-bilinear map $f : M \times N \to L$, there exists a unique R-linear $h : M \otimes N \to L$, making the diagram



commute.

Proof. Uniqueness is clear, since we must have that

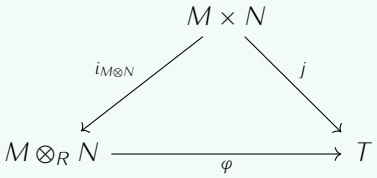
$$h(m \otimes n) = f(m, n)$$

since the pure tensors generate, h must be unique, if it exists. Therefore, suffices to show the above extends to an R-linear map $M \otimes_R N \to L$. This follows from the map

$$\begin{aligned} R^{\oplus(M\times N)} &\to L \\ e_{(m,n)} &\mapsto f(m, n) \end{aligned}$$

extending to a linear map (by the universal property of the direct sum), and that this map vanishes on $\mathcal{K}$. Therefore, h extends to $M \otimes_R N$ from the pure tensors. □

Proposition 2.3. Let M, N be R-modules, T an R-module, $j : M \times N \to T$ an R-bilinear map, (T, j) satisfying the universal property of tensors. Then there exists a unique R-linear isomorphism $\varphi : M \otimes N \to T$, such that



commutes.

Proof. By the universal property of tensor product, such a map φ exists, with $\varphi(m \otimes n) = j(m, n)$. Similarly, we have a homomorphism $\psi : T \to M \otimes_R N$. In particular,

$$\psi \circ \varphi \circ i_{M\otimes N} = i_{M\otimes N} = \mathrm{id}_{M\otimes N} \circ i_{M\otimes N}$$

In particular, by uniqueness in the universal property, we must have that $\psi \circ \varphi = \mathrm{id}_{M\otimes N}$. □

Lecture 3

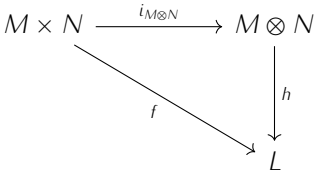
Proposition 2.4. Suppose M, N are R-modules, then

$$\sum_i m_i \otimes n_i = 0 \in M \otimes_R N$$

if and only if for all R-modules L, and every R-bilinear map $f : M \times N \to L$ has

$$\sum_i f(m_i, n_i) = 0$$

Proof. Suppose $\sum m_i \otimes n_i = 0$, let $f : M \times N \to L$ be bilinear. Then f factors through $M \times N \to M \otimes_R N$, and we can write



In this case, we have that

$$\sum_i f(m_i, n_i) = \sum_i h(i(m, n)) = \sum_i h(m_i \otimes n_i) = h\left(\sum_i m_i \otimes n_i\right) = h(0) = 0$$

Conversly, if

$$\sum_i m_i \otimes n_i \neq 0$$

then by definition,

$$\sum_i i_{m\otimes n}(m_i, n_i) \neq 0$$

□

Example 2.5

Let k be a field, and consider the tensor product

$$k^m \otimes k^\ell$$

Suppose k^m has basis $\{e_1, \ldots, e_m\}$, and k^ℓ has basis $\{f_1, \ldots, f_\ell\}$, then

$$k^m \otimes k^\ell = \operatorname{span}_k\{v \otimes w \mid v \in k^m, w \in k^\ell\} = \operatorname{span}_k\left\{e_i \otimes f_j\right\}$$

Claim 2.6. $\{e_i \otimes f_j\}$ is a basis.

Proof. Suppose we have

$$\sum_{ij} \alpha_{ij}(e_i \otimes f_j) = 0$$

For every $1 \leq a \leq m, 1 \leq b \leq \ell$, define a bilinear map

$$T_{ab} : k^m \times k^\ell \to k$$
$$T_{ab}((v_i), (w_j)) = v_a w_b$$

This is a k-bilinear map. By proposition 2.4,

$$0 = \sum_{i,j} \alpha_{ij} T_{ab}(e_i, f_j) = \sum_{i,j} \alpha_{ij} \delta_{ia} \delta_{jb} = \alpha_{ab}$$

□

Example 2.7

More concretely, let us consider $\mathbb{R}^2 \otimes \mathbb{R}^2$. We have a basis of size 4, given by

$$e_1 \otimes f_1, e_1 \otimes f_2, e_2 \otimes f_1, e_2 \otimes f_2$$

What do pure tensors look like?

$$(\alpha e_1 + \beta e_2) \otimes (\gamma f_1 + \delta f_2) = \alpha\gamma(e_1 \otimes f_1) + \alpha\delta(e_1 \otimes f_2) + \beta\gamma(e_2 \otimes f_1) + \beta\delta(e_2 \otimes f_2)$$

These are not generic elements of $\mathbb{R}^2 \otimes \mathbb{R}^2$, since the vectors

$$(\alpha\gamma, \alpha\delta) \quad \text{and} \quad (\beta\gamma, \beta\delta)$$

are linearly dependent. In particular,

$$e_1 \otimes f_1 + 2e_1 \otimes f_2 + 3e_2 \otimes f_1 + 4e_2 \otimes f_2$$

is *not* a pure tensor.

Example 2.8 (warning)

First consider

$$\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

In this case,

$$2 \otimes 1 = 1 \otimes 2 = 1 \otimes 0 = 0$$

Now consider

$$2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2$$

But in this case,

$$2 \otimes 1 \neq 0$$

since we can define a bilinear map

$$B : 2\mathbb{Z} \times \mathbb{Z}/2 \to \mathbb{Z}/2$$
$$B(2m, x) = mx$$

In this case,

$$B(2, 1) = 1 \cdot 1 = 1 \neq 0$$

However, if $M' \leq M, N' \leq N$ are submodules, and

$$\sum_i m_i \otimes n_i = 0$$

in $M' \otimes N'$, then

$$\sum_i m_i \otimes n_i = 0$$

in $M \otimes N$.

Proposition 2.9. If

$$\sum m_i \otimes n_i = 0 \in M \otimes_R N$$

then there are finitely generated R-submodules $M' \leq M, N' \leq N$, such that

$$\sum m_i \otimes n_i = 0 \in M' \otimes_R N'$$

Intuitively, a proof that the sum is zero is finite, and so it can only involve finitely many expressions. We

can take them to be the generators.

Proof.

$$\sum m_i \otimes n_i 0 =\in M \otimes N = \frac{R^{\oplus(M\times N)}}{\mathcal{K}}$$

then

$$\sum_i e_{(m_i,n_i)} = 0 \in \mathcal{K}$$

This means that we can write the left hand side as a finite sum of the generators of $\mathcal{K}$. Taking all the elements of M and N which appear, gives the result. □

Corollary 2.10. Let A, B be torsion-free abelian groups, then $A \otimes_{\mathbb{Z}} B$ is torsion free.

Proof. Suppose

$$n \cdot \left(\sum_i a_i \otimes b_i \right) = 0 \in A \otimes B$$

for some $n \geq 1$. By proposition 2.9, there exists finitely generated subgroups $A' \leq A, B' \leq B$, such that

$$n \cdot \left(\sum_i a_i \otimes b_i \right) = 0 \in A' \otimes B'$$

By the structure theorem of finitely generated abelian groups, $A' \cong \mathbb{Z}^r, B' \cong \mathbb{Z}^s$, and so we have that

$$A' \otimes B' \cong \mathbb{Z}^{rs}$$

which is torsion free. Contradiction. □

Example 2.11

$$\mathbb{C}^2 \otimes_{\mathbb{C}} \mathbb{C}^3 \cong \mathbb{C}^6$$

as $\mathbb{C}$-vector spaces, and we also have that $\mathbb{C}^6 \cong \mathbb{R}^1 2$ as $\mathbb{R}$-vector spaces. On the other hand,

$$\mathbb{C}^2 \otimes_{\mathbb{R}} \mathbb{C}^3 \cong \mathbb{R}^4 \otimes_{\mathbb{R}} \mathbb{R}^6 \cong \mathbb{R}^{24}$$

Proposition 2.12.

1. $M \otimes N \cong N \otimes N$
2. $(M \otimes N) \otimes P \cong M \otimes (N \otimes P) \cong M \otimes N \otimes P$, where we define $M \otimes N \otimes P$ using trilinear maps.
3. $\left(\bigoplus_i M_i\right) \otimes P \cong \bigoplus_i (M_i \otimes P)$
4. $R \otimes_R M \cong M$,

Proof. See examples sheet 1. □

Example 2.13

Using proposition 2.12, we can compute

$$\begin{aligned} R^m \otimes R^\ell &\cong \left(\oplus_{i=1}^m R\right) \otimes \left(\oplus_{j=1}^\ell R\right) \\ &\cong \oplus_{i,j} R \\ &\cong R^{m\ell} \end{aligned}$$

2.1 Tensor product of R-linear maps

Proposition 2.14. For R-linear maps $f : M \to M', g : N \to N'$, then there exists a unique R-linear map

$$f \otimes g : M \otimes N \to M' \otimes N'$$

with

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

Proof. Uniqueness is clear since the pure tensors generate. For existence, we can use the universal property on the R-bilinear map

$$\begin{aligned} T : M \times N &\to M' \otimes N' \\ T(m, n) &= f(m) \otimes g(n) \end{aligned}$$

□

Lecture 4

Exercise: $(f \otimes g) \circ (h \otimes i) = (f \circ h) \otimes (h \circ i)$.

Example 2.15

Let $T : k^a \to k^c$ and $S : k^b \to k^d$ be linear. Then

$$(T \otimes S)(e_i \otimes e_j) = T(e_i) \otimes S(e_j) = \sum_{\ell,t} [T]_{\ell i}[S]_{tj}(f_\ell \otimes f_t)$$

where $[T]$ is the matrix representation of T. If we order the basis of $k^a \otimes k^b$ by

$$e_1 \otimes e_1, \ldots, e_1 \otimes e_c, e_2 \otimes e_1, \ldots, e_2 \otimes e_c, \ldots, e_a \otimes e_c$$

and a similar ordering for the range, then

$$[T \otimes S] = \begin{pmatrix} [T]_{11}S & \cdots & [T]_{1a}S \\ \vdots & \ddots & \vdots \\ [T]_{c1}S & \cdots & [T]_{ca}S \end{pmatrix}$$

is the *Kronecker product* of $[T]$ and $[S]$.

Proposition 2.16. Let $f : M \to M', g : N \to N'$ be $\mathbb{R}$-linear.

(i) If f, g are isomorphisms, then so is $f \otimes g$,

(ii) if f and g are surjective, so is $f \otimes g$.

Proof. For (i), $(f^{-1} \otimes g^{-1}) = (f \otimes g)^{-1}$, since we have that $(f \otimes g) \circ (h \otimes i) = (f \circ h) \otimes (h \circ i)$.
For (ii), notice that $\mathrm{im}(f \otimes g)$ contains all pure tensors in $M' \otimes N'$. □

Example 2.17

If $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = pn$, then we have

$$(f \otimes \mathrm{id}) : \mathbb{Z} \otimes \mathbb{Z}/p \to \mathbb{Z} \otimes \mathbb{Z}/p$$

is the zero map, as

$$(f \otimes \mathrm{id})(a \otimes b) = (pa) \otimes b = a \otimes (pb) = a \otimes 0 = 0$$

But $\mathbb{Z} \otimes \mathbb{Z}/p \cong \mathbb{Z}/p$ which is nonzero.

2.2 Tensor product of algebras

Let B, C be R-algebras. Then we have $B \otimes_R C$ as an R-module. We would like to define the multiplication by

$$(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$$

This is well-defined. Fix $(b, c) \in B \times \mathbb{C}$, then we have a bilinear map

$$\begin{aligned} B \times C &\to B \otimes C \\ (b', c') &\mapsto (bb') \otimes (cc') \end{aligned}$$

which gives us a map $B \otimes C \to B' \otimes C'$, with

$$b' \otimes c' \mapsto (bb') \otimes (cc')$$

It is easy to show that this then satisfies the ring axioms. Hence $B \otimes C$ is a ring. The R-algebra structure will be given by

$$\begin{aligned} R &\to B \otimes C \\ r &\mapsto (r1_B) \otimes 1_C = r(1_B \otimes 1_C) = 1_B \otimes (r1_C) \end{aligned}$$

Example 2.18

There is an isomorphism

$$\varphi : R[x_1, \ldots, x_n] \otimes_R R[t_1, \ldots, t_r] \cong R[x_1, \ldots, x_n, t_1, \ldots, t_r]$$

Proof. We have an R-basis for the left hand side, which is

$$x^k \otimes t^\ell$$

and we also have a R-basis for the right hand side,

$$x^k t^\ell$$

Define

$$\varphi(x^k \otimes t^\ell) = x^k t^\ell$$

which gives us a R-module isomorphism. Moreover,

$$\varphi(r \otimes 1) = r1 = 1$$

and by distributivity, suffices to show

$$\varphi((x^k \otimes t^\ell)(x^m \otimes t^n)) = x^k t^\ell x^m t^n$$

which is clear by definition. □

More generally,

$$\frac{R[x_1, \ldots, x_n]}{I} \otimes \frac{R[t_1, \ldots, t_r]}{J} \cong \frac{R[x_1, \ldots, x_n, t_1, \ldots, t_r]}{L} \cong \frac{R[x_1, \ldots, x_n, t_1, \ldots, t_r]}{I^e + J^e}$$

where $I^e = \langle I \rangle \trianglelefteq R[x_1, \ldots, x_n, t_1, \ldots, t_r]$ denotes the extension of I.

Example 2.19

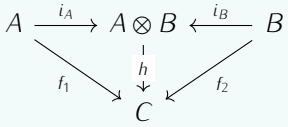
$\frac{\mathbb{C}[x,y,z]}{\langle f,g\rangle} \otimes \frac{\mathbb{C}[w,u]}{h}$ is isomorphic as $\mathbb{C}$-algebras to

$$\frac{\mathbb{C}[x,y,z,w,u]}{\langle f,g,h\rangle}$$

Proposition 2.20 (universal property of tensor product of algebras). Let A, B be R-algebras, for every R-algebra C, and R-algebra homomorphisms $f_1 : A \to C$ and $f_2 : B \to C$, there exists a unique R-algebra map

$$h : A \otimes B \to C$$

such that



commutes, where $i_A(a) = a \otimes 1$, $i_B(b) = 1 \otimes b$. Moreover, this characterises $(A \otimes B, i_A, i_B)$ uniquely (up to isomorphism).

Proof. $A \otimes B$ is generated, as an R-algebra, by

$$\{a \otimes 1 \mid a \in A\} \cup \{1 \otimes b \mid b \in B\}$$

This then implies the uniqueness of h, as it defines h on the generators. For the existence, define the bilinear map $A \times B \to C$, given by

$$f(a,b) = f_1(a)f_2(b)$$

Using the universal property of tensor product of modules, there exists $h : A \otimes B \to C$ which is R-linear, with

$$h(a \otimes b) = f_1(a)f_2(b)$$

It is then easy to show that h is an algebra homomorphism. □

Consider $R[x_1, \dots, x_n, t_1, \dots, t_r]$ from above. We have natural embeddings from $R[x_1, \dots, x_n]$ and $R[t_1, \dots, t_n]$. Given f_1, f_2 as above, we see that the image of the x_i is determined by f_1, and the image of t_i is determined by f_2. Therefore,

$$R[x_1, \dots, x_n, t_1, \dots, t_r] \cong \mathbb{R}[x_1, \dots, x_n] \otimes R[t_1, \dots, t_r]$$

as it satisfies the universal property.

Lecture 5

If we have $f : A \to A', g : B \to B'$ which are algebra homomorphisms, then the tensor product of R-linear maps,

$$f \otimes g : A \otimes B \to A' \otimes B'$$

is an R-algebra homomorphism. Moreover, we have R-algebra isomorphisms

- $(R/I) \otimes (R/J) \cong R/(I + J)$
- $A \otimes B \cong B \otimes A$,
- $(A \otimes B) \otimes C \cong A \otimes (B \otimes C)$,
- $A \otimes B^n \cong (A \otimes B)^n$,

2.3 Restriction and extension of scalars

Restriction of scalars

We will have a ring homomorphisms $f : R \to S$, let M be an S-module, so M is also an R-module,

$$r \cdot m := f(r)m$$

for $r \in R, m \in M$. The fact that this is a module is clear by our definition, since it is just the composition

$$R \xrightarrow{f} S \longrightarrow \mathrm{End}(M)$$

Example 2.21

If we consider the embedding $\mathbb{R} \hookrightarrow \mathbb{C}$, then $\mathbb{C}^n$ is a C-vector space, but also an $\mathbb{R}$-vector space, of dimension $2n$.

Extension of scalars

Let $f : R \to S$ be a ring homomorphism, M be an S-module (thus an R-module by restriction of scalars), N is an R-module. From this, we can form

$$M \otimes_R N$$

which is an R-module. In fact, $M \otimes_R N$ is also an S-module, with

$$s \cdot (m \otimes n) := (sm) \otimes n$$

Is this well defined? We have an R-bilinear map

$$\begin{aligned} M \times N &\to M \otimes_R N \\ (m, n) &\mapsto (sm) \otimes n \end{aligned}$$

By the universal property, we have a map

$$h_s : M \otimes_R N \to M \otimes_R N$$

which is R-linear, and $h_s(m \otimes n) = (sm) \otimes n$. Now define

$$\begin{aligned} \varphi : S &\to \mathrm{End}(M \otimes_R N) \\ \varphi(s) &= h_s \end{aligned}$$

Which is a ring homomorphism, and so, we have an S-module structure on $M \otimes_R N$.

Example 2.22

We know from before that $S \otimes_R R \cong S$ as R-module, with

$$s \otimes r \mapsto s \cdot f(r)$$

But in fact, this is also S-linear, since

$$s' \cdot (s \otimes r) = (s's) \otimes r \mapsto s's \cdot f(r)$$

For example, this implies that

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R} \cong \mathbb{C}$$

as $\mathbb{C}$-vector spaces.

Example 2.23

If M is an S-module, N_i are R-modules, then

$$M \otimes_R \left(\bigoplus_i N_i \right) \cong \bigoplus_i (M \otimes_R N_i)$$

as S-modules.

In this case,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^n \cong \mathbb{C}^n$$

as $\mathbb{C}$-vector spaces.

Example 2.24

Consider $\mathbb{C}^n$ as a $\mathbb{C}$-module. Restricting to $\mathbb{R}$,

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

as $\mathbb{R}$-vector spaces. Now extending scalars,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^{2n} \cong \mathbb{C}^{2n}$$

as $\mathbb{C}$-vector spaces.

Example 2.25

Now consider $\mathbb{R}^n$ as an $\mathbb{R}$-vector space. Extending scalars,

$$\mathbb{R}^n \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}^n$$

over $\mathbb{C}$. Restricting to $\mathbb{R}$,

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

Example 2.26

Consider $\mathbb{Z}^n$ as an $\mathbb{Z}$-module, and let $f : \mathbb{Z} \to \mathbb{Z}/2$ be the quotient map. Extending scalars,

$$(\mathbb{Z}/2) \otimes_{\mathbb{Z}} \mathbb{Z}^n \cong (\mathbb{Z}/2)^n$$

Example 2.27

Consider

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell$$

One way to compute this:

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong_{\mathbb{R}} \mathbb{R}^{2n} \otimes \mathbb{R}^\ell \cong_{\mathbb{R}} \mathbb{R}^{2n\ell} \cong_{\mathbb{R}} \mathbb{C}^{n\ell}$$

where $\cong_{\mathbb{R}}$ denotes isomorphism as $\mathbb{R}$-vector spaces. Another way to do this:

$$\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \cong_{\mathbb{C}} \mathbb{C}^n \otimes_{\mathbb{C}} \left(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^\ell\right) \cong_{\mathbb{C}} \mathbb{C}^n \otimes \mathbb{C}^\ell \cong_{\mathbb{C}} \mathbb{C}^{n\ell}$$

The first isomorphism is given by

$$v \otimes u \mapsto v \otimes (1 \otimes u)$$

Combining these, the isomorphism $\mathbb{C}^n \otimes_{\mathbb{R}} \mathbb{R}^\ell \to \mathbb{C}^n \otimes \mathbb{C}^\ell$ sends

$$v \otimes u \mapsto v \otimes u$$

where we use the inclusion $\mathbb{R}^\ell \hookrightarrow \mathbb{C}^\ell$.

Proposition 2.28. Let M be an S-module, N be an R-module, then

$$M \otimes_R N \cong M \otimes_S (S \otimes_R N)$$

as S-modules. In particular, the isomorphism is given by

$$m \otimes n \mapsto m \otimes (1 \otimes n)$$
$$(sm) \otimes n \mapsfrom m \otimes (s \otimes n)$$

Proposition 2.29. Let M, M' be S-modules, N, N' be R-modules, then we have S-module isomorphisms

(i) $M \otimes_R N \cong N \otimes_R M$, via $m \otimes n \to n \otimes m$

(ii) $(M \otimes_R N) \otimes_R N' \cong M \otimes_R (N \otimes_R N')$

(iii) $(M \otimes_R N) \otimes_S M' \cong M \otimes_S (N \otimes_R M')$

(iv) $M \otimes_R (\bigoplus_i N_i) \cong \bigoplus_i (M \otimes_R N_i)$

Proof. We will prove (iii). Using proposition 2.28, we have

$$\begin{aligned}(M \otimes_R N) \otimes_S M' &\cong (M \otimes_S (N \otimes_R S)) \otimes_S M' \\ &\cong M \otimes_S \left((N \otimes_R S) \otimes_S M'\right) \\ &\cong M \otimes_S (N \otimes_R M')\end{aligned}$$

□

Example 2.30

As $\mathbb{C}$-vector spaces,

$$\mathbb{C} \otimes_{\mathbb{R}} (\mathbb{R}^\ell \otimes_{\mathbb{R}} \mathbb{R}^k) \cong (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^\ell) \otimes_{\mathbb{C}} (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}^k) \cong \mathbb{C}^\ell \otimes \mathbb{C}^k \cong \mathbb{C}^{\ell k}$$

Corollary 2.31. If N, N' are R-modules, then

$$S \otimes_R (N \otimes N') \cong_S (S \otimes_R N) \otimes_S (S \otimes N')$$

Proof. By proposition 2.28 and proposition 2.29 (ii):

$$S \otimes_R (N \otimes_{\mathbb{R}} N') \cong (S \otimes_R N) \otimes_R N' \cong (S \otimes_R N) \otimes_S (S \otimes_R N')$$

□

Lecture 6

By induction, we have that

$$S \otimes_R (N_1 \otimes_R \cdots \otimes_R N_\ell) \cong (S \otimes_R N_1) \otimes_S \cdots \otimes_S (S \otimes_R N_1)$$

Extension of scalars for morphisms

Let $f : N \to N'$ be R-linear, where N, N' are R-modules, M is an S-module. Then we have a map

$$\mathrm{id} \otimes f : M \otimes_R N \to M \otimes_R N'$$

In particular, it is S-linear, as

$$(\mathrm{id} \otimes f)(s(m \otimes n)) = (\mathrm{id} \otimes f)((sm) \otimes n) = (sm) \otimes f(n) = s(m \otimes f(n)) = s(\mathrm{id} \otimes f)(m \otimes n)$$

Given $T : \mathbb{R}^n \to \mathbb{R}^\ell$ which is an $\mathbb{R}$-linear map, $\mathbb{R}^n$ with basis $e_1, \ldots, e_n$ and $\mathbb{R}^\ell$ with basis $f_1, \ldots, f_\ell$. In this case, consider

$$\mathrm{id} \otimes T : \mathbb{C} \otimes \mathbb{R}^n \to \mathbb{C} \otimes \mathbb{R}^\ell$$

Note that $\mathbb{C} \otimes \mathbb{R}^n$ has basis $1 \otimes e_1, \ldots, 1 \otimes e_n$. In particular,

$$(\mathrm{id} \otimes T)(1 \otimes e_i) = 1 \otimes T(e_i) = 1 \otimes \sum_{j=1}^{\ell} T_{ji} f_j = \sum_{j=1}^{\ell} T_{ji}(1 \otimes f_j)$$

Thus, T and $\mathrm{id} \otimes T$ have the same matrix representation.

Extension of scalars of algebras

Let A, B be R-algebras. Recall that in this case, $A \otimes_R B$ is also an R-algebra. In fact, $A \otimes_R B$ is an A-algebra (and by symmetry a B-algebra). For example, we have

$$A \to A \otimes_R B$$
$$a \mapsto a \otimes 1$$

Example 2.32

$S \otimes_R R[x_1, \ldots, x_n] \cong_S S[x_1, \ldots, x_n]$ (where $\cong_S$ denotes isomorphism of S-algebras).

Proof. We already have an S-module isomorphism

$$\varphi : S \otimes_R R[x_1, \ldots, x_n] \to S[x_1, \ldots, x_n]$$

with $\varphi(s \otimes f) = sf$. It is easy to show that

$$\varphi(s \otimes 1) = s$$

and that φ preserves multiplication. □

More generally, we have that

$$S \otimes \left(\frac{R[x_1, \ldots, x_n]}{I} \right) \cong \frac{S[x_1, \ldots, x_n]}{I^e}$$

where $I^e = \langle f(I) \rangle$ is the ideal generated by I under the ring homomorphism $f : R \to S$.

Proposition 2.33. Suppose A is an R-algebra, B is an S-algebra, then $A \otimes_R B$ is an S-algebra. Moreover,

$$A \otimes_R B \cong_{S\text{-alg}} (A \otimes_R S) \otimes B$$

Proof. $A \otimes_R B$ is a B-algebra, and we can then restrict scalats to S. The isomorphism is clear from the module case, as all we need to check is it preserves multiplication. □

Proposition 2.34. Suppose A, B are R-algebras, then

$$S \otimes_R (A \otimes_R B) \cong_{S\text{-alg}} (S \otimes_R A) \otimes_S (S \otimes_R B)$$

2.4 Exactness properties of the tensor product

Let M be a fixed R-module. Define

$$\mathrm{T}_M(N) = M \otimes_R N$$

where N is an R-module. If $f : N \to N'$ is R-linear, then we have an induced map

$$\mathrm{T}_M(f) = \mathrm{id}_M \otimes f : \mathrm{T}_M(N) \to \mathrm{T}_M(N')$$

Suppose we have an exact sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

of R-modules. We will show that we have an exact sequence

$$\mathrm{T}_M(A) \xrightarrow{\mathrm{T}_M(f)} \mathrm{T}_M(B) \xrightarrow{\mathrm{T}_M(g)} \mathrm{T}_M(C) \longrightarrow 0$$

That is, T_M is a *right exact functor* from R-modules to R-modules.

Definition 2.35 (Hom)
Suppose Q, P are R-modules, then we can define

$$\mathrm{Hom}_R(Q,P) = \{f : Q \to P \mid f \text{ is } R\text{-linear}\}$$

This is an R-module itself, with

$$(r \cdot \varphi)(q) = r \cdot \varphi(q)$$

Definition 2.36 (Hom functors)
We have two functors,

1. $\mathrm{Hom}_R(Q, \cdot)$, where Q is a fixed R-module,
2. $\mathrm{Hom}_R(\cdot, P)$, where P is a fixed R-module.

Suppose we have $f : N \to N'$ which is R-linear, then the action on morphisms are

$$\begin{aligned}\mathrm{Hom}_R(Q,f) : \mathrm{Hom}_R(Q,N) &\to \mathrm{Hom}_R(Q,N') \\ \varphi &\mapsto f \circ \varphi =: f_*(\varphi)\end{aligned}$$

On the other hand, $\mathrm{Hom}_R(\cdot, P)$ is contravariant. That is,

$$\begin{aligned}\mathrm{Hom}_R(f,P) : \mathrm{Hom}_R(N',P) &\to \mathrm{Hom}_R(N,P) \\ \varphi &\mapsto \varphi \circ f =: f^*(\varphi)\end{aligned}$$

Proposition 2.37 (left exactness of the Hom-functors).
1. If

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is exact, then so is

$$0 \longrightarrow \mathrm{Hom}_R(Q,A) \xrightarrow{\mathrm{Hom}_R(Q,f)} \mathrm{Hom}_R(Q,B) \xrightarrow{\mathrm{Hom}_R(Q,g)} \mathrm{Hom}_R(Q,C)$$

2. If

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact, then so is

$$0 \longrightarrow \mathrm{Hom}_R(C,P) \xrightarrow{\mathrm{Hom}_R(g,P)} \mathrm{Hom}_R(B,P) \xrightarrow{\mathrm{Hom}_R(f,P)} \mathrm{Hom}_R(A,P)$$

In both cases, we say that the respective Hom functor is *left exact*.

Proof. Omitted. □

Lemma 2.38. Consider a (not necessarily exact) sequence

$$A \xrightarrow{f} B \xrightarrow{g} C$$

and suppose for all R-module P, the sequence

$$\mathrm{Hom}_R(C,P) \longrightarrow \mathrm{Hom}_R(B,P) \longrightarrow \mathrm{Hom}_R(A,P)$$

is exact, then the original sequence is exact.

Proof. **Step 1:** let $P = C$. Then we get the sequence

$$\mathrm{Hom}_R(C, C) \longrightarrow \mathrm{Hom}_R(B, C) \longrightarrow \mathrm{Hom}_R(A, C)$$

which is exact by assumption. Under this,

$$\mathrm{id}_C \mapsto \mathrm{id}_C \circ g = g \mapsto g \circ f$$

Thus, we have that $g \circ f = 0$, and so $\mathrm{im}(f) \subseteq \ker(g)$.

Step 2: Let $P = \mathrm{coker}\, f = \frac{B}{\mathrm{im}(f)}$. In this case, we have

$$\mathrm{Hom}(C, \mathrm{coker}(f)) \longrightarrow \mathrm{Hom}(B, \mathrm{coker}(f)) \longrightarrow \mathrm{Hom}(A, \mathrm{coker}(f))$$

Let $h : B \to \mathrm{coker}(f)$ denote the quotient map. Then $h \circ f = 0$, and so by exactness, there exists $e : C \to \mathrm{coker}(f)$, with

$$\mathrm{Hom}(g, \mathrm{coker}(f))(e) = e \circ g = h$$

In particular, $\ker(g) \subseteq \ker(h) = \mathrm{im}(f)$. □

Recall that we have a bijection $\mathrm{Hom}_R(M \otimes_R N, L) \cong \mathrm{Bil}(M \times N, L)$ from the universal property of the tensor product. But

$$\mathrm{Bil}(M \times N, L) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, L))$$

and so we have an isomorphism

$$\mathrm{Hom}_R(M \otimes N, L) \cong \mathrm{Hom}_R(N, \mathrm{Hom}_R(M, L))$$

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